Subject Code: KCE070

PAPER ID-310214

Roll No: | | | | | | | | | | | | | |

BTECH
(SEM VII) THEORY EXAMINATION 2024-25
RAILWAY, WATERWAY AND AIRWAY ENGINEERING

TIME: 3 HRS **M.MARKS: 100**

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A

1. Attempt *all* questions in brief. **2 x 10 = 20**

Q no.	Question	CO	Level
a.	What is the function of ballast in railway tracks?	CO1	K2
b.	What is sleeper density?	CO1	K2
c.	Define a marshalling yard.	CO2	K2
d.	What is cant excess?	CO2	K2
e.	What is a railway signal?	CO3	K3
f.	Define ballastless track.	CO3	K2
g.	What is the purpose of an airport hangar?	CO4	K2
h.	Discuss different types of runway lighting systems.	CO4	K3
i.	Define littoral drift.	CO5	K2
j.	What is sounding in water transport?	CO5	K3

SECTION B

2. Attempt any *three* of the following: **10 x 3 = 20**

Q no.	Question	CO	Level
a.	Explain the concept of coning of wheels and its significance in railway operations.	CO1	K3
b.	What are the different types of railway stations, and how are they classified based on their functions?	CO2	K3
c.	What are the key design considerations for high-speed rail tracks to ensure safety and stability at higher speeds?	CO3	K3
d.	Why is drainage management critical in airport design and operation?	CO4	K4
e.	What is a port? Discuss the requirements of a good port.	CO5	K3

SECTION C

3. Attempt any *one* part of the following: **10 x 1 = 10**

Q no.	Question	CO	Level
a.	Define a permanent way and explain its importance in railway infrastructure.	CO1	K3
b.	Explain the types of locomotives used in Indian Railways and their respective advantages.	CO1	K3

4. Attempt any *one* part of the following: **10 x 1 = 10**

Q no.	Question	CO	Level
a.	A broad-gauge railway track has a curve with a radius of 1500 meters. The maximum sanctioned speed is 120 km/h, and the equilibrium speed is 90 km/h. Calculate the following: • The required super-elevation. • The maximum permissible speed.	CO2	K4

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b.	Describe the types of crossings used in railway tracks and their applications.	CO2	K3

5. Attempt any *one* part of the following: 10 x 1 = 10

Q no.	Question	CO	Level
a.	What is track interlocking? Discuss the different types of interlocking systems used in railway operations.	CO3	K3
b.	What is the Centralized Train Control (CTC) system, and how does it improve train operations and safety	CO3	K4

6. Attempt any *one* part of the following: 10 x 1 = 10

Q no.	Question	CO	Level
a.	Explain the role of zoning laws in ensuring the safety of airport operations.	CO4	K3
b.	What are the major components of an airport, and how do they function together?	CO4	K3

7. Attempt any *one* part of the following: 10 x 1 = 10

Q no.	Question	CO	Level
a.	Why is tidal analysis crucial for port and harbor operations?	CO5	K3
b.	Discuss the factors to be considered when selecting a location for a harbor.	CO5	K4

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